

BIF 501 MCQ FOR MID TERM
Subjective

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Regards VUWAYS Team

Welcome To vuways Study Help

Bif501 solved papers

Paper 1

ReadSeq 2 Ans:

READSEQ is a useful sequence converter developed by D.G.Gilbert at Indiana University, USA

- Recognizes DNA or Protein sequence file and interconvert them.

Q

EcoR1 restriction fragments 2 Ans:

EcoRI (Escherichia coli)

GAATTC

EcoRI (pronounced "eco R one") is a restriction endonuclease enzyme isolated from species E. coli. The *Eco* part of the enzyme's name originates from the species from which it was isolated, while the *R* represents the particular strain, in this case RY13. The last part of its name, the *I*, denotes that it was the first enzyme isolated from this strain. *EcoRI* is a restriction enzyme that cleaves DNA double helices into fragments at specific sites. It is also a part of the restriction modification system.

Q

TSS predictions 5 Ans:

Translation Start Site (TSS)

Translation starts with ATG that codes for methionine in a polypeptide. Translation

Start Site

(TSS)

TSS Prediction

GCCATGGCGA ...

ACGATGCTGT ...

GACATGGTAC ...

AGGATGGGCT ...

Assumption

Certain nucleotides prefer to be around TSS than others

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MOAZZAM

- The biased nucleotide distribution information is a basis for translation start prediction.

Coding Potential

Hexamer frequencies in coding versus non-coding regions may provide important insights

Frequency of X(A,G,C,T) at position i is

F_i

$$(X) = \log(C_i)$$

$$(X)/N_i(X)$$

Example

Which one is more probable to be a Translation Start?

CACC ATA GC TCGA

ATG TT

Solution

We can use frequency table and the scoring function as under;

$$S_i = \log(F_i)$$

$$(X)/0.25)$$

We can call it Information Content (IC).

Algorithm

- Build a mathematical model, based on collected translation start sequence
- For each candidate translation start sequence, apply the model and get a score
- If the score is larger than zero, predict it is a translation start; the higher score, the higher the probability the prediction is true.

Q

Proteome comparison and advantages:

Ans:

Comparative Proteomics

Introduction

- For a new Genome, predicted genes are translated
- The collection of Protein sequences encoded by the genome makes up the proteome of that individual. All against all, self comparison
- Comparison of all proteins with each other within proteome
- Identify unique proteins from the ones having paralogs
- Identify Gene families

Cluster Analysis

- To sort out the relationships of all related proteins
- Clustering classify the proteins based on some objective criteria e.g
 - E value cut off
 - Distance in alignment

Clustering by subgraph

- Each sequence is a vertex
- Significant alignment score is an edge
- Trimming by removing weak edges (High P/E) Clustering by Linkage
- Each sequence is a vertex
- Significant alignment score is an edge
- Trimming by removing weak edges (High P/E)
- Or remove $> e^{-6}$
- Remaining subgraph should share 2/3rd of edges

Single Linkage

- A group of sequences in all-against-all comparison is subjected to MSA
- Create distance matrix
- Neighbour joining is then used to do clustering.

Core Proteome

- All-against-all comparison provides an indication of Gene/Protein families
- Unique set of proteins is core proteome. Conclusions
- Genome is translated into proteome
- Self comparison of proteome yields gene families and duplications
- Unique set of proteins is core proteome Significance
- Helps finding orthologs, Gene families and domains
- Proteins with highly significant alignment score are likely to be orthologs
- Mostly the proteins related to core biological functions.

Q

Effects of deletion of nucleotide 1, 2,3 in detail 5marks Ans:

Introduce deletions in DNA-dramatically alters its protein product. Deleting three consecutive nucleotides results in minor changes in the protein.

The phrase

THE SLY FOX AND THE SHY DOG

(written in triplets)

Turns into nonsense after deleting one letter Y from SLY

THE SYF OXA NDT HES HYD OG or

two letters LY from SLY

THE SFO XAN DTH ESH YDO G

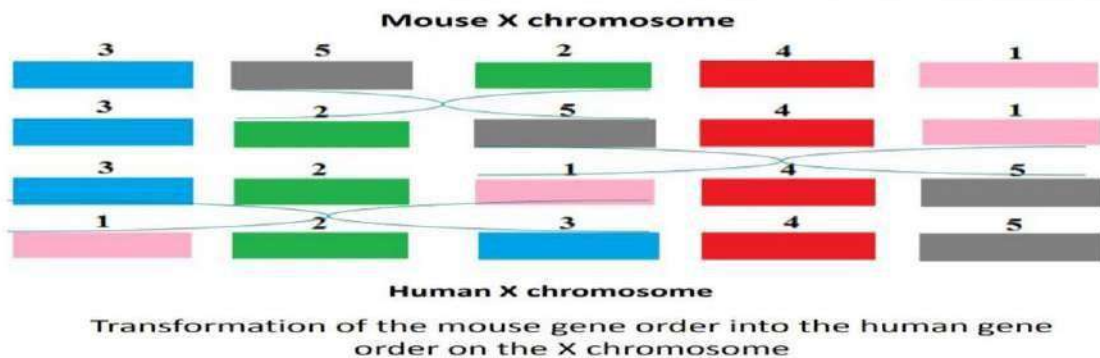
but makes some sense after deleting three nucleotides SLY

THE SOX AND THE SHY DOG

Q

Waardenburg's syndrome Hearing loss Two Different colored eyes Gene present on chromosome 2 Splotch gene in mice. Human genome-mouse genome Cut into 300 genomic fragments-Synteny blocks Chromosome 2 in humans mouse chromosomes 1, 2, 3, 5, 6, 7, 10, 11, 12, 14, and 17. Genome rearrangement results in a change of gene ordering Analysis of human and mouse genomes-250 genomic rearrangements.

Brute Force Algorithm



Q.

Xml:

XML Format

- Extensible Markup Language
- Readable by both man and machine
- Becoming standard data format for transferring genome data

GENSCAN 2 marks Ans:

GENSCAN developed by Chris Burge and Samuel Karlin. GENSCAN combines coding region and splicing signal predictions into a single framework. Splice site prediction-coding region appear on one side of the site Such statistics are used in the HMM framework of GENSCAN that merges splicing site statistics, coding region statistics, and motifs near the start of the gene. The accuracy of GENSCAN decreases for genes with many short exons or with unusual codon usage.

Q

Goal of biological databases 3

Ans

Biological databases in general store biological data and their main goals are

- Data storage
- Information retrieval
- Knowledge discovery

Q:

Bypass algorithm 3:

Branch and Bound Algorithm

NEXTVERTEX Algorithm

BYPASS Algorithm

Skip the subtree rooted at vertex (a, i)

Increment a_i (unless $a_i = k$)

BYPASS(a, i, L, k)

1. for j i to 1
2. if $a_j < k$
3. $a_j = a_j + 1$
4. return (a, j)

5. return (a, 0)

Q

What are long common subsequence? Write purpose.2

The simplest form of a sequence similarity analysis is the Longest Common Subsequence (LCS) problem, where we eliminate the operation of substitution and allow only insertions and deletions. A subsequence of a string v is simply an (ordered) sequence of characters (not necessarily consecutive) from v .

Q

Prediction of Exons Ans:

Introduction

Exons can be predicted within ORF from the information gathered about

- splice junctions

Approach

For each segment [acceptor, donor], we get three scores (coding potential, donor score, acceptor score)

Various possibilities

- all three scores are high – probably true exon
- all three scores are low – probably not a real exon
- all in the middle – ?
- some scores are high and some are low – ??

Prediction

- Collect a set of exons and non-exons
- Score them using our scoring schemes
- Plot them as follows
- 管draw策 a separating line between exons and non-exons

Linear Discriminate Analysis linear discriminate analysis (LDA) finds an optimal plane surface that best separates points that belong to two classes

- For example, if there are ten true exons and ten introns, and two feature
- These samples could be represented by 20 points in a two-dimensional space
- LDA would compute a straight line through the space that can best separate the two classes with the minimal classification error

Conclusion

- Collect a set of exons and non-exons
- Score them using our scoring schemes
- Plot them as follows
- 管draw策 a separating line between exons and non-exons.

Q

What is Edit distance? give its example. 5 Ans:

Edit distance between two strings as the minimum number of editing operations needed to transform one string into another, where the edit operations are insertion, deletion, and substitution of one symbol for another. It is often the case that the i th symbol in one sequence corresponds to a symbol at different position in other. Mutation in DNA-evolutionary process: DNA replication- substitutions, insertions, and deletions of nucleotides, leads to edited DNA texts. Whether the i th symbol in one DNA sequence corresponds to the i th symbol in the other

Edit Distance

A T A T A T A T -
- T A T A T A T A

align the $(i+1)$ -st letter in ATATATAT against the i th letter in TATATATA for $1 \leq i \leq 7$

1	2	3	4	5	6	7	8
A	T	A	T	A	T	A	T
-	T	A	T	A	-	A	T
	1	2	3	4		5	6

Edit Distance

TGCATAT can be transformed into **ATCCGAT** with five editing operations

TGCATAT

↓
TGCATA

delete last T

↓
TGCAT

delete last A

↓
ATGCAT

insert A at front

↓
AT**CC**AT

substitute C for G at position 3

↓
ATCCGAT

insert a G before the last A

Edit distance between TGCATAT and ATCCGAT is at most 5

Edit Distance

TGCATAT can also be transformed into **ATCCGAT** with four editing operations

TGCATAT

↓
ATGCATAT

insert A at front

↓
ATGCAAT

delete T at position 6

↓
ATGCGAT

substitute G for A at position 5

↓
ATCCGAT

substitute C for G at position 3

Edit distance between TGCATAT and ATCCGAT is 4

Q

Write the classes of biological database.3

Classification

Biological databases can be classified as

- Primary databases
- Secondary databases
- Specialized Databases

Classification

Biological databases can also be classified on the bases of types of data as

- Nucleotide databases
- Protein databases
- RNA databases
- Genome databases
- Expression databases

Q

What is the power of DNA comparison?5 Ans:

Discovery of new gene-no idea of Functions

Find similarities with genes of known function. Newly discovered cancer-causing - sis oncogene matched a normal gene involved in growth and development called platelet-derived growth factor. Oncogene v-sis is the simian sarcoma virus. Scientists became suspicious that cancer might be caused by a normal growth gene. Discovery of cystic fibrosis gene. Abnormal secretions, and is diagnosed in children produce abnormally thick mucus that clogs the lungs. 10 million Americans are carriers of the cystic fibrosis gene. There is a 25% chance that the child will have cystic fibrosis.

In 1989, 1 million nucleotides on the chromosome 7. The area around the cystic fibrosis gene was sequenced. Database of all known genes. Similarities between a gene that had already been discovered, code for adenosine triphosphate (ATP)

binding proteins. These proteins constitute the ion transport channel. The disease involves sweat secretions with abnormally high sodium content. Link between cancer-causing genes and normal growth genes and elucidating the nature of cystic fibrosis. Q

what are the conserved pattern of DNA?5

Profiles

32 nucleotide
7 sequences

```

1- CGGGGCTGGGGTCGTCACATTCCCCTTTTCGATA
2- TTTGAGGGGTGCCCAATAACCAAAGCGGACAAA
3- GGGATGCCGTTTGACGACCTAAATCAACGGCC
4- AAGGCCAGGAGCGCCTTTGCTGGGTTCTACCTG
5- AATTTTCTAAAAAAGATTATAATGTCGGGTCCTC
6- CTGCTGTACAACTGAGATCATGCTGCTTCAAC
7- TACATGATCTTTTGTGGATGAGGGAATGATGC
  
```

Figure 1

Profiles

$$P = \text{ATGCAACT}$$

$$l = 8$$

- 1- CGGGGCTATGCAACTGGGGTCGTCACATTCCCCTTTTCGATA
- 2- TTTGAGGGGTGCCCAATAAAATGCAACTCCAAAGCGGACAAA
- 3- GGATGCAACTGATGCCGTTTGACGACCTAAATCAACGGCC
- 4- AAGGATGCAACTCCAGGAGCGCCTTTGCTGGTTCTACCTG
- 5- AATTTTCTAAAAAAGATTATAATGTCGGTCCATGCAACTTC
- 6- CTGCTGTACAACTGAGATCATGCTGCATGCAACTTTCAAC
- 7- TACATGATCTTTTGATGCAACTTGGATGAGGGAATGATGC

Figure 2

Profiles

$$P = \text{ATGCAACT}$$

$$l = 8$$

- 1- CGGGGCTATGCAACTGGGGTCGTCACATTCCCCTTTTCGATA
- 2- TTTGAGGGGTGCCCAATAAAATGCAACTCCAAAGCGGACAAA
- 3- GGATGCAACTGATGCCGTTTGACGACCTAAATCAACGGCC
- 4- AAGGATGCAACTCCAGGAGCGCCTTTGCTGGTTCTACCTG
- 5- AATTTTCTAAAAAAGATTATAATGTCGGTCCATGCAACTTC
- 6- CTGCTGTACAACTGAGATCATGCTGCATGCAACTTTCAAC
- 7- TACATGATCTTTTGATGCAACTTGGATGAGGGAATGATGC

Figure 3

Profiles

$$P = \text{ATGCAACT}$$

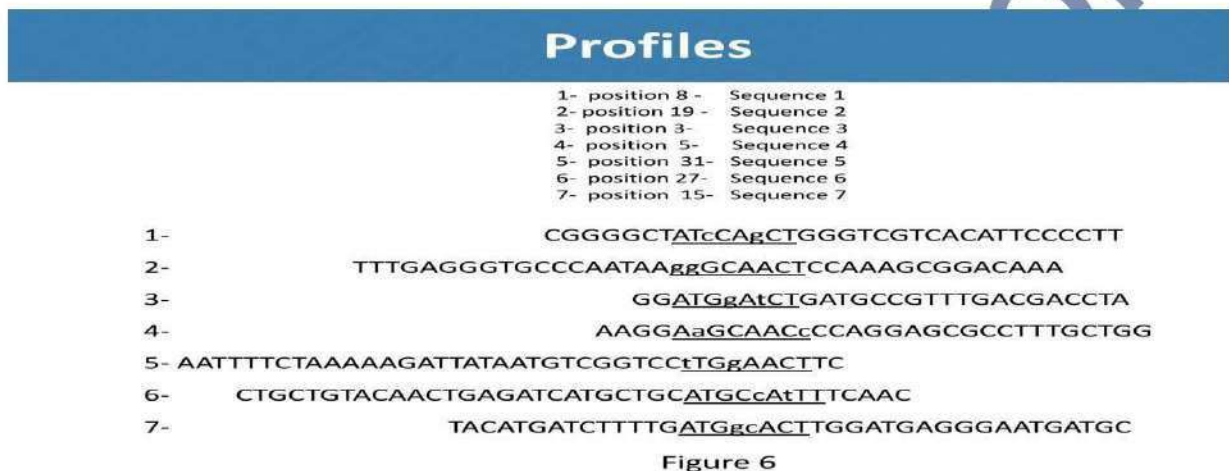
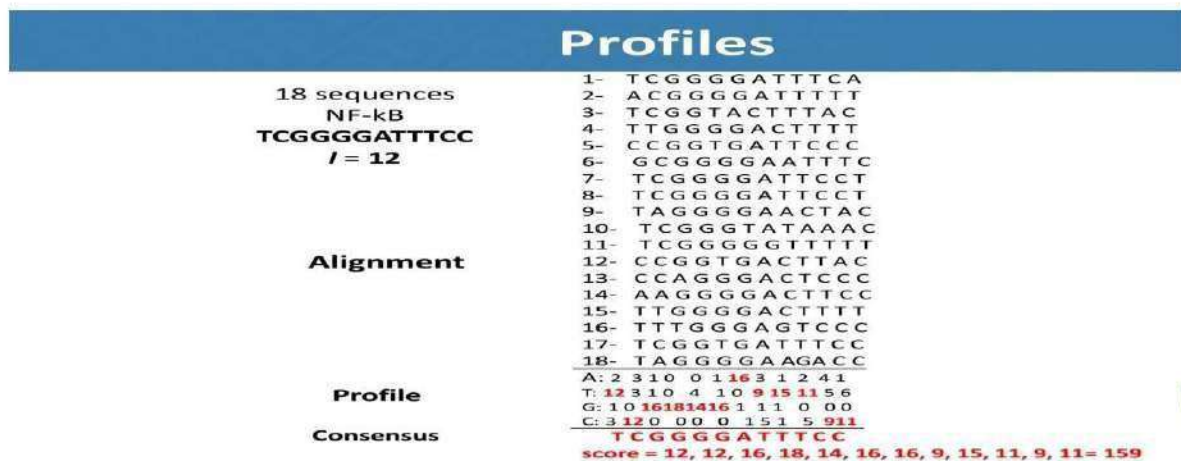
$$l = 8$$

$$7 \times (32 + 8) = 280 \text{ nucleotides}$$

$$\text{Probability} = 280/4^8 = 0.004$$

- 1- CGGGGCTATcCagCTGGGGTCGTCACATTCCCCTTTTCGATA
- 2- TTTGAGGGGTGCCCAATAAaggGCAACTCCAAAGCGGACAAA
- 3- GGATGgAtCTGATGCCGTTTGACGACCTAAATCAACGGCC
- 4- AAGGAaGCAACcCCAGGAGCGCCTTTGCTGGTTCTACCTG
- 5- AATTTTCTAAAAAAGATTATAATGTCGGTCCtTGgAACTTC
- 6- CTGCTGTACAACTGAGATCATGCTGCATGcCAtTTTCAAC
- 7- TACATGATCTTTTGATGgcACTTGGATGAGGGAATGATGC

Figure 4



Q

1. Define structure annotation. What we can do by using this technique.

Structural Annotation

Identification of gene features like;

- Promoters
- Terminators
- Shine-Dalgarno sites
- DNA motifs
- Co-transcription units
- Operons in microbes

Q

Enzyme commission 5:

Enzyme Commission (EC) numbers

Enzyme Commission (EC) under IUBMB

Enzymes are classified on the basis of reactions they catalyze. EC a.b.c.d a one of 6 classes of biochemical reactions b group of substrate c acceptor molecule d details of biochemical reaction. tripeptide aminopeptidases

EC 3.4.11.4

3 Hydrolases (use water to break substrate)

3.4 Hydrolases that act on peptide bonds

3.4.11 Hydrolases that cleave amino terminal amino acids of polypeptide tripeptide aminopeptidases

EC 3.4.11.4

Hydrolases that cleave amino terminal amino acids of a tripeptide .70% of e.coli genes shared a and b, catalize same biochemical reaction.

Q

1. Describe synteny of human x chromosome. (2) Synteny

- Arrangement of genes on the chromosomes
- Comparing gene orders provides a good measure of similarities and differences among different organisms.

Gene Order (Synteny)

Human Chromosomes

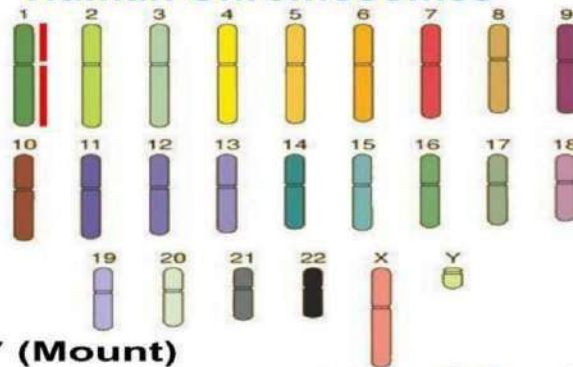


Figure 10.7 (Mount)

2. Write note on Magpie and genequiz. (2)

MAGPIE (Multipurpose Automated Genome Project Investigation Environment)

It's an automated genome analysis tool that is used for structural annotation.

GeneQuiz

Focuses on deriving a predicted protein function based upon available evidence, including evaluation of similarity to the closest homologue in database.

3. WRITE ANY SIX APPLICATIONS OF BIOINFORMATICS. (3)

Drug Development Crop Improvement

Microbial Genome

Gene Therapy

Biotechnology

Comparative Study

Evolutionary Studies

Veterinary Science

Q

5. Write application of bioinformatics in molecular medicine and biotechnology. (5) Ans:

Molecular Medicine:

1. The human genome project
2. 3000-4000 hereditary disease including Cystic Fibrosis and Huntingtons disease
3. Response to an environmental stress causes cancers, heart disease, diabetes
4. Human Genome Project Data Base Biotechnology:
 1. Archaeoglobus fulgidus and Thermotoga maritima
 2. Corynebacterium glutamicum
 3. Xanthomonas campestris
 4. Lactococcus lactis

Q

Synteny of mice chromosomes:

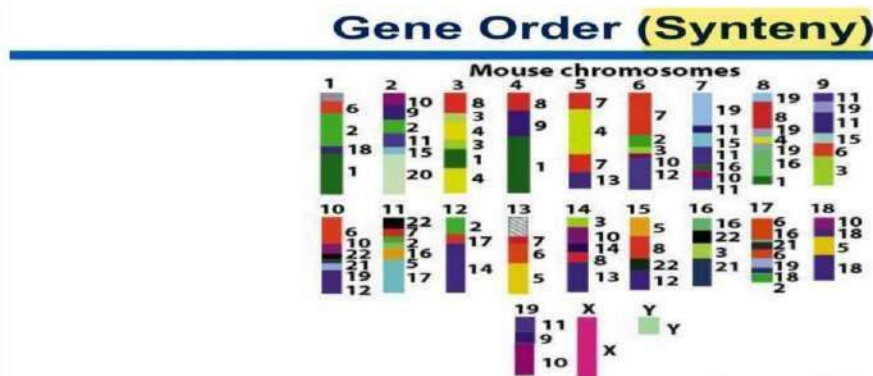


Figure 10.7 (Mount)

Q.matches,mismatches, insertion, deletions.

Ans

Columns that contain the same letter in both rows are called matches, while columns containing different letters are called mismatches. The columns of the alignment containing one space are called indels, with the columns containing a space in the top row called insertions and the columns with a space in the bottom row deletions.

Q

Gene therapy;

Gene therapy-used to treat, cure or even prevent disease.clinical trials.

(Gene therapy is the therapeutic delivery of nucleic acid into a patient's cells as a drug to treat disease.[1] The first attempt at modifying human DNA was performed in 1980 by Martin Cline, but the first successful nuclear gene transfer in humans, approved by the National Institutes of Health, was performed in May 1989.[2] The first therapeutic use of gene transfer as well as the first direct insertion of human DNA into the nuclear genome was performed by French Anderson in a trial starting in September 1990.

Between 1989 and February 2016, over 2,300 clinical trials had been conducted, more than half of them in phase I.[3]

Not all medical procedures that introduce alterations to a patient's genetic makeup can be considered gene therapy. Bone marrow transplantation and organ transplants in general have been found to introduce foreign DNA into patients.[4] Gene therapy is defined by the precision of the procedure and the intention of direct therapeutic effects.)

Q.heterochromatin and euchromatin. Staining with dyes

- Dense heterochromatin
- Light Euchromatin
- Heterochromatic regions are poorly transcribed (expressed)
- Euchromatic regions are highly expressed

Q.

Sequence LOGOS:

Ans.

A visual representation of a position-specific distribution. Easy for nucleotides, but we need colour to depict up to 20 amino acid proportions. Overall height at position is proportional to the information content.

- Proportions of each nucleotide/amino acid are in relation to their observed frequency, with most frequent on top, next most frequent below.

Q.

Gene Expression Omnibus (GEO)

- A public repository for the archiving and distribution of gene expression data submitted by the scientific community.
- MIAME compliant data

– Minimum Information About a Microarray

Experiment <http://www.mged.org/Workgroups/MIAME/miame.html>

- Convenient for deposition of gene expression data, as required by funding agencies and journals
- Curated resource for gene expression data
- Browsing, querying, analysis and retrieval GEO has four kinds of records
- Sample (GSM) preparation and description of the samples
- Platform (GPL) technology used and the features detected – microarray or RNASeq
- Series (GSE) defines a set of samples and how they are related
- Datasets (GDS) sample data collections assembled by GEO.

Q

Dynamic

Programming

- Dynamic programming is used to find an optimal alignment of two sequences and its scores.
- It is a method by which a larger problem may be solved by first solving smaller, partial versions of the problem.
- Three steps in dynamic programming:
 - Initialization
 - Matrix filling (scoring)
 - Traceback (alignment)
- Initialization: Create matrix with M+1 columns and N+1 rows where M and N correspond to the size of sequences to be aligned
- Matrix filling:
Fill the matrix with highest possible score
- Traceback:

Move from the last corner and follow the arrow

Q.local alignment problems:

Biological significant in certain parts of DNA fragments, maximize the alignment score $s(v_i, w_j)$

... v_i, w_j over all substrings $v_i \dots v_i$ of v and $w_j \dots w_j$ of w . Local Alignment Problem- not extend over entire length as in Global Alignment Problem.

Local Alignment Problem:

Find the best local alignment between two strings

Input: Strings v and w and a scoring matrix δ Output:

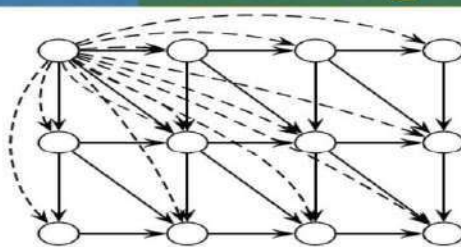
Substring of v and w whose global

alignment, as defined by δ , is maximal among all global alignment of all substrings of v and w .

Global Sequence Alignment- finding the longest path between vertices $(0,0)$ and (n,m) in the edit graph. Local Alignment-finding the longest path among the paths between arbitrary vertices (i, j) and (i', j') in the edit graph. Find the longest path between every pair of vertices (i, j)

and (i', j') then select longest of these computed paths. Instead of finding the longest path from (i, j) to (i', j') , LAP- finding the longest path from the source $(0,0)$ to every other vertex by adding edges to weight 0.

Local Alignment Problem



These edges make the source vertex $(0,0)$ a predecessor provide a "free ride" from the source to any other vertex (i, j) . Following recurrence reflects the transformation of the edit graph

$$s_{i,j} = \max \begin{cases} 0 \\ s_{i-1,j} + \delta(v_i, -) \\ s_{i,j-1} + \delta(-, w_j) \\ s_{i-1,j-1} + \delta(v_i, w_j) \end{cases}$$

Q.

Ans

Endosymbiont hypothesis

- originally proposed in 1883 by Andreas Schimper, but extended by Lynn Margulis in the 1980s
- Mitochondria and Chloroplast are derived from endosymbiotic bacteria.

(The endosymbiotic hypothesis concerns the origin of mitochondria and chloroplasts, two organelles contained within various eukaryotic cells. According to this hypothesis, these organelles originated as separate prokaryotic organisms that were taken inside a primordial eukaryotic cell. Such symbiotic relationships in which two species are dependent upon one another to varying extents served as crucial elements of the evolutionary progression of eukaryotic cells. The endosymbiotic hypothesis has developed greatly over the past century. Continued research on this topic has resulted in a considerable amount of evidence and support for this hypothesis, often referred to now as the endosymbiotic theory. With this research, the endosymbiotic hypothesis has been extended to include the evolutionary significant concept of secondary endosymbiosis, which adds to the complexity of the

evolutionary process. The true significance of the endosymbiotic hypothesis becomes poignantly evident, however, when actually relating the scientific basis of this concept to actual examples of endosymbionts in nature.)

Q.how microbes are useful in climate changed study?

Ans

Climate change Studies

Study the genomes of microbes that use carbon dioxide as their sole carbon source. Q.

Difference between introns and exons,

Ans

Eukaryotic genes are complicated

- They possess exons which are protein coding regions and are interrupted with introns. Whole gene is transcribed as a large transcript
- Introns are spliced leaving an ORF that is translated

Introns start on the 5' ends as GT and end as AG towards 3' end GT-----AG

- This trend is highly conserved

Q

enlist edit operation? 2 Ans. the edit operations are insertion, deletion, and substitution of one symbol for another. Q

Note on orf? Ans:

Open reading frames (ORFs) Fenome of length n as a sequence of $n/3$ codons. The three stop codons, (TAA, TAG, and TGA) The subsegments of these that start from a start codon, ATG, are ORFs. ORFs within a single genomic sequence may overlap since there are six possible reading frames: three on one strand starting at positions 1, 2, and 3, and three on the reverse strand.

(In molecular genetics, an open reading frame (ORF) is the part of a reading frame that has the potential to be translated. An ORF is a continuous stretch of codons that contain a start codon (usually AUG) and a stop codon (usually UAA, UAG or UGA). [1] An ATG codon within the ORF (not necessarily the first) may indicate where translation starts. The transcription termination site is located after the ORF, beyond the translation stop codon. If transcription were to cease before the stop codon, an incomplete protein would be made during translation. [2] In eukaryotic genes with multiple exons, ORFs span intron/exon regions, which may be spliced together after transcription of the ORF to yield the final mRNA for protein translation.)

Q

Bug theory:

Ans

"The Gold Bug" by Edgar Allan provided some clue of finding DNA motifs, one of the character find parchment written below

53++!305))6*;**48**26)4+.)4+);806*;**48**!8'60)
)85;]8*:+*8!83(88)5*!;46(;88*96*?;8)*+;
485);5*!2:*+(;4956*2(5*-)8'8*;4069285)
 ;)6!8) 4+++;1(+9;**48**081;8:8+1;**48**!85;4)485!
 528806*81(+9;**48**;(88;4(+?34;**48**)4+;161;;
 188;+?;

“;48” codes for **“THE”**

53++!305))6*THE26)H+.)H+)TE06*THE!E'60)
)E5T]E*:+*E!E3(EE)5*!TH6(TEE*96*?TE)*
 +(THE5) T5*!2:*+(TH956*2(5*-H)E'E*T
 H0692E5)T)6!E)H++T1(+9THE0E1TE:E+1THE!
 E5TH)HE5!52EE06*E1(+9THET(EETH(+?3HT
 HE) H+T161T:1EET+?T

“;” for **“T”**
“4” for **“H”**
“8” for **“E”**

Q

Data retrieval:

Ans

Data Retrieval

- Nearly all biological databases are available for download as simple text (flat) files
- A local version of the database allows one greater freedom in processing the data

Entrez

- is an integrated search engine which allows users to search and retrieve different data from the NCBI
- It can be accessed from the site www.ncbi.nlm.nih.gov/Entrez/.
- integrates PubMed and 39 other scientific literatures, nucleotide and protein databases
- protein domain data, population studies, expression data, pathways, genome details and taxonomic information.

Q

Bypass algorithms: Ans: